

Internal energy and The First Law

Internal energy: Energy not kinetic or potential

- 1) thermal energy (heat)
- 2) mechanical/strain energy
- 3) Chemical energy
- 4) Magnetization

Note: everything below is spatial = Eulerian

Internal energy $\mathcal{U}$ of Ω_t : $\mathcal{U}[\Omega_t] = \int_{\Omega_t} \rho u \, dv_x$

$u(x,t)$ = internal energy density / unit mass

First Law of Thermodynamics

$$\boxed{\frac{d}{dt} \mathcal{U}[\Omega_t] = Q[\Omega_t] + W[\Omega_t]} \quad \text{for all } \Omega_t \in \mathcal{B}_t$$

Net heating: $Q[\Omega_t] = \text{rate of thermal energy gain/loss}$

Net working: $W[\Omega_t] = \text{rate of mechanical energy gain/loss}$

Unit of heating/working/power: Watt = $W = \frac{J}{s}$

Net heating

2 forms of heat gain/loss:

I) Body heating: $Q_b[\Omega_t] = \int_{\Omega_t} \rho H dV$

H = heat gain/power per unit mass $\left[\frac{W}{kg}\right]$ $W = \frac{J}{s}$

Example: radiogenic heating

$$H = H_0 e^{-\lambda t}$$

H_0 = initial power per unit mass $\left[\frac{W}{kg}\right]$

λ = decay constant $\left[\frac{1}{s}\right]$

Note: $A = \rho H$ power per unit volume $\left[\frac{W}{m^3}\right]$

II) Surface heating: $Q_s[\Omega_t] = - \int_{\partial\Omega_t} \mathbf{q} \cdot \hat{\mathbf{n}} dA$

$\mathbf{q}(\mathbf{x}, t)$ = heat flux vector $\left[\frac{W}{m^2}\right]$

Fourier's law: $\mathbf{q} = -\kappa \nabla T$ (Lecture 10)

κ = thermal conductivity $\left[\frac{W}{mK} = \frac{J}{msK}\right]$

Net heating is the rate at which heat is added or lost from Ω_t

$$Q[\Omega_t] = Q_b[\Omega_t] + Q_s[\Omega_t] = \int_{\Omega_t} \rho H dV - \int_{\partial\Omega_t} \mathbf{q} \cdot \hat{\mathbf{n}} dA$$

Energy balance with net heating

First law with $W=0$: $\frac{d}{dt} \mathcal{U}[\Omega_t] = Q[\Omega_t]$

where $\mathcal{U}[\Omega_t] = \int_{\Omega_t} \rho u \, dV_x$

$$Q[\Omega_t] = \int_{\Omega_t} \rho H \, dV_x - \int_{\partial\Omega_t} \mathbf{q} \cdot \underline{n} \, dA_x$$

substituting:

$$\frac{d}{dt} \int_{\Omega_t} \rho u \, dV_x = - \int_{\partial\Omega_t} \mathbf{q} \cdot \underline{n} \, dA_x + \int_{\Omega_t} \rho H \, dV_x$$

using derivative relative to mass and divergence Thm

$$\int_{\Omega} (\rho \dot{u} + \nabla_x \cdot \mathbf{q} + \rho H) \, dV_x$$

by the arbitrary ness of Ω_t we have

$$\boxed{\rho \dot{u} + \nabla \cdot \mathbf{q} = \rho H} \quad \text{local Eulerian form}$$

Following lin. mom. balance $\rightarrow$ conservative form

$$\boxed{\frac{\partial}{\partial t} (\rho u) + \nabla \cdot [\underline{v} \rho u + \mathbf{q}] = \rho H} \quad \Rightarrow \text{HW 9}$$

- still in terms of internal energy density !

Relate u to measurable quantities

Fundamental thermodynamic relation: $U = U(S, V)$

$$\boxed{dU = T dS - p dV} \quad \text{thermo book version}$$

U = internal energy [J]

S = entropy [$\frac{J}{K}$] V = volume [m^3]

T = temperature [K] p = pressure [Pa]

$\Rightarrow$ good for global statements (whole system)

Continuum mechanics is local

$u = \frac{U}{m}$ = specific internal energy [$\frac{J}{kg}$]
energy/mass m = mass

need $u = u(T, p)$

$$\boxed{du = c_p dT - \frac{\alpha T}{\rho} dp} \rightarrow \text{take a good thermo class.}$$

(Legendre transformations & Maxwell relations)

c_p = specific heat capacity [$\frac{J}{kg K}$]

ρ = mass density [$\frac{kg}{m^3}$]

$\alpha = -\frac{1}{\rho} \left. \frac{\partial \rho}{\partial T} \right|_p$ = isobaric thermal expansivity [$\frac{1}{K}$]

Advection - Conduction Equation

Energy balance: $\frac{\partial}{\partial t}(\rho u) + \nabla \cdot [\underline{v} \rho u + \underline{q}] = \rho r$

$\underline{v} \rightarrow$ momentum balance

ρ, r parameters (given)

$u, \underline{q}$ unknown variables

1 equation with 2 unknowns

$\Rightarrow$ need to relate u and $\underline{q}$

Assume: $p = \text{const.}$ and $c_p = \text{const.}$

Thermodynamics: $du = c_p dT$ ($dp = 0$)

$$u = u_0 + c_p (T - T_0) \quad c_p = \text{const.}$$

Fouriers law: $\underline{q} = -\kappa \nabla T$ (Lecture 10)

physical parameters:

1) $c_p = \text{specific heat capacity}$ $\left[\frac{\text{J}}{\text{kg K}} \right]$

2) $\kappa = \text{thermal conductivity}$ $\left[\frac{\text{W}}{\text{m K}} \right]$

Assume: 1) properties are constant (ρ, c_p, κ)

2) flow is incompressible $\nabla \cdot \underline{v} = 0$

substitute and simplify

$$\rho c_p \frac{\partial T}{\partial t} - \nabla \cdot [\underline{v} \rho c_p T - \kappa \nabla T] = \rho r \quad \Rightarrow \text{HW9}$$

Advective heat flux: $q_A = \underline{v} \rho c_p T$

Conductive heat flux: $q_c = -\kappa \nabla T$

Heat production: ρr

$\Rightarrow$ Advection - Diffusion - Reaction Equation

Diffusion $\sim$ Conduction

Reaction $\sim$ Production

In limit of $\underline{v} \rightarrow 0$

$$\frac{\partial T}{\partial t} - D \nabla^2 T = \frac{r}{c_p} \quad \underline{\text{Heat Equation}}$$

where

$\nabla^2 = \nabla \cdot \nabla = \text{Laplacian}$ (Lecture 10)

$D = \frac{\kappa}{\rho c_p}$ thermal diffusivity $[\frac{m^2}{s}]$

$\kappa \rightarrow$ heat flux

$D \rightarrow$ change in T

Example: Steady State Geotherm

Steady state: $\frac{\partial T}{\partial t} = 0 \Rightarrow -\kappa \nabla^2 T = \rho r$

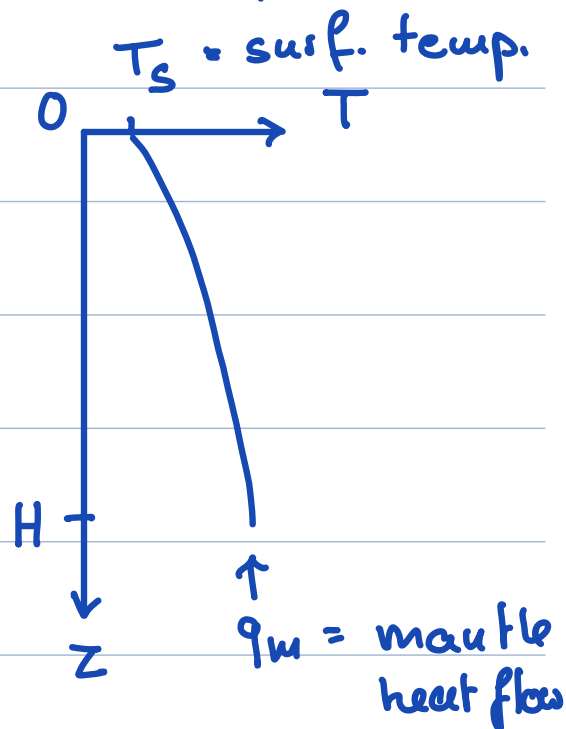
One dimension: $-\kappa \frac{d^2 T}{dz^2} = \rho r$

Boundary Conditions:

$$T(z=0) = T_s$$

$$q(z=H) = -\kappa \left. \frac{dT}{dz} \right|_{z=H} = -q_m$$

↑
upward!



Integrate once: $-\kappa \frac{dT}{dz^2} = \rho r \quad -\kappa \frac{dT}{dz} = \rho r z + c_1$

Use bottom BC: $-\kappa \left. \frac{dT}{dz} \right|_{z=H} = \rho r H + c_1 = -q_m$
 $\Rightarrow c_1 = -q_m - \rho r H$

substitute c_1 : $-\kappa \frac{dT}{dz} = \rho r z - \rho r H - q_m$

$\Rightarrow$ heat flow: $q(z) = -\rho r (H - z) - q_m$

absence of heat flow: $q(z) = -q_m$

Integrate again: $-\kappa T = \rho r \frac{1}{2} z^2 - (\rho r H + q_m) z + c_2$

Use top BC: $-\kappa T_s = c_2$

$$\Rightarrow -\kappa T = \frac{\rho r}{2} z^2 - (\rho r H + q_m) z - \kappa T_s$$

$$\Rightarrow \text{geotherm: } T(z) = T_s + \left(\frac{\rho r H}{\kappa} + \frac{q_m}{\kappa} \right) z - \frac{\rho r}{2\kappa} z^2$$

Chicxulub (Espinosa-Cardena et al. 2016)

$A = \rho r = 1.5 \mu\text{W}/\text{m}^3$ vol. heat production

$\kappa = 2.67 \frac{\text{W}}{\text{mK}}$ thermal conductivity

$H = 35 \text{ km}$

$T_s = 26.5^\circ\text{C}$ (Merida, Mexico)

$q_m = 28 \text{ mW}/\text{m}^2$

$$\Rightarrow T(H) = 707^\circ\text{C} \quad q(0) = 79 \text{ mW}/\text{m}^2$$

1) Granite melts at 650°C (amphibolite at 800°C)

$\Rightarrow$ base of crust is close to melting

2) Mean heat flow of continents is $65 \text{ mW}/\text{m}^2$

$\Rightarrow$ increased heat flow 66 Myrs after impact